



Fundamentals of ELECTRO-OPTIC SYSTEMS DESIGN

COMMUNICATIONS, LIDAR, AND IMAGING

Sherman Karp and Larry B. Stotts

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Fundamentals of Electro-Optic Systems Design

Communications, Lidar, and Imaging

Using fundamentals of communication theory, thermodynamics, information theory and propagation theory, this book explains the universal principles underlying a diverse range of electro-optical systems. From visible / infra-red imaging, to free space optical communications and laser remote sensing, the authors relate key concepts in science and device engineering to practical systems issues.

A broad spectrum of coherent and incoherent electro-optical systems is considered, accompanied by many real-world examples. The authors also present new insights into the fundamental limitations of these systems when operating through deleterious channels.

Accompanied by online examples of processed images and ideas, this uniquely tailored guide to the fundamental principles underlying modern electro-optical systems is an essential reference for all practicing engineers, graduate students, and academic researchers in optical engineering.

Sherman Karp received his Ph.D. from the University of Southern California, and has gone on to work with NASA, and as Principal Scientist at DARPA. The author of several books, he has also been awarded the SECDEF Medal for Meritorious Civilian Service, and NOSC “Scientist of the Year.” He is a Fellow of the IEEE.

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“With the completion of *Fundamentals of Electro-Optic Systems Design*, Sherman Karp and Larry B. Stotts have created a single comprehensive book for anyone having anything to do with the vast field of electro-optics. The detailed systems design principles, examples, charts, graphs, and methods bring together under one cover the information to handle the applications given by the subtitle *Communications, Lidar, and Imaging*. The basic theories and their relationship to real-world hardware constraints such as noise and scattering are covered in full detail with necessary citations to decades of electro-optics research. From a systems design point-of-view, Karp and Stotts blend Lidar, laser communications, and imaging into a logical path to analyze, design, and test complex electro-optics. The communication chapters covering modulation, coding, and propagation in various media are not found anywhere else unless one wades through thousands of research papers and reports. If you are a scientist or engineer who has to manipulate photons, *Fundamentals of Electro-Optic Systems Design* belongs on your bookshelf – near the front.”

Robert K. Tyson, The University of North Carolina at Charlotte

“This book uniquely treats electro-optical system design from an engineering viewpoint emphasizing real world applications and where theory works and does not work. These perspectives make this book a must-have reference for the scientist or engineer involved with electro-optical system design.”

Tony Tether, Former DARPA Director 2001 to 2009

“*Fundamentals of Electro-Optic Systems Design* is a comprehensive and authoritative treatment of free-space optical communications and Lidar. Topics range from diffraction, photoelectric detection, effects of scattering and optical turbulence, and even signal coding, modulation and error correction.”

Joseph W. Goodman, Stanford University

“The book is written by very knowledgeable and very experienced individuals in the field of electro-optical systems. Their writing and explanations make the material very accessible. It is clear and well presented.”

Ronald Phillips, University of Central Florida

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Communications, Lidar, and Imaging

SHERMAN KARP
LARRY B. STOTTS



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In Memorandum

The authors dedicate this book to our esteemed friend and colleague, Professor Robert M. Gagliardi, who recently passed away. We have known Bob for almost fifty years, and we will miss him. He was truly a great and gentle man.

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Preface

The authors have been active participants in the area of electro-optic systems for over four decades, covering the introduction of laser systems and low loss optical fibers and the institutionalizing of photonic systems into everyday life. Yet for all the literature that exists, and all the work that has been accomplished, we felt that no single book existed that integrated the entire field of electro-optics, reaching back to all the fundamental building blocks and providing enough examples to be useful to practicing engineers. After much discussion and a slow start, we decided first to reference as much material as possible, bringing forth only the highlights necessary to guide researchers in the field. Then we decided to minimize mathematical developments by relegating them, as much as possible, to explanatory examples. What has evolved in our development is a clear statement of the duality of time and space in electro-optic systems. This had been touched upon in our earlier work, but has been brought forth clearly in this book in the duality of modulation index in time, and contrast in space. In doing so, and in other areas, we feel that this book contains new material with regard to the processing of spatial images which have propagated through deleterious channels. We feel that this book contains much new material in the areas of communications and imaging through deleterious channels.

In [Chapter 1](#), we reach back to the true foundations of modern physics, the establishment of the first two laws of thermodynamics. While taken for granted, it is the first law that explains why we can see stars at the edge of the universe, and governs the radiant properties of propagating systems. The second law and the insight of Claude Shannon have created the modern field of Information Theory. Using his fundamental definitions of channel capacity we are able to establish the duality of time and space in electro-optics. This requires one basic mathematical development that is included in Appendix A, and is developed in [Chapters 3](#) and [4](#).

In [Chapter 2](#), we present the development of Maxwell's equations, which we correlate with radiant properties of propagating fields. We bring this development to the point where we can establish the conditions under which duality applies. In the process we demonstrate how the radiant properties can be used to make engineering calculations in both temporal and spatial systems, passing from the transmitting optics through to the detector.

In [Chapter 3](#), we focus on the behavior of the photo-detection system. Using Appendix A, we introduce the duality between temporal systems and spatial systems. We identify the information-bearing portion of the received signal (as defined by Shannon) and the modulation index of incoherent communication systems.

In [Chapter 4](#), we delve more deeply into spatial systems and the effects on imaging. Again using Appendix A, we show the duality between the modulation index in temporal systems and contrast in spatial systems. This is related to the classic definition and merely extends it based upon the newer understanding of the behavior of electro-optic systems. We then proceed further in digital communication systems and show how we develop optimum systems under a few different constraints.

In [Chapter 5](#), we investigate in detail the concepts of contrast and visibility and their relationship to signal-to-noise ratio. We then discuss the duality of modulation index in communication theory and contrast in imaging. We also introduce the concept of sub-visibility imaging, which allows a direct trade-off between improved contrast and deteriorated signal-to-noise ratio. Imagery demonstrating this trade-off is presented, and applications are discussed.

In [Chapter 6](#), we review various optical modulation schemes used in optical communications. This will include a discussion of spectral efficiency and the energy per bit per unilateral noise density, parameters used extensively in RF communications, but just coming into use in optical communications. These latter aspects help us determine the optimal means for optical communications, especially when erbium-doped fiber amplifiers are involved.

In [Chapter 7](#), we provide the basics on block forward error correction (FEC) encodings, and highlight the arguably most popular FEC, Reed-Solomon codes. We then will note other important types of FEC schemes that can be employed, based on the authors' bias. This material is not meant to be an exhaustive survey of FEC coding, but rather included in this book to give the reader a basic background and knowledge of FEC to illustrate their utility in optical communications today. Application of FEC to optical systems will be discussed in a subsequent chapter.

In [Chapter 8](#), we discuss some of the key aspects of the signal modulation and coding schemes used in fiber optic and free-space optical communications systems today. Most notably, we will review the use of return-to-zero and non-return-to-zero in coding the information streams and see their effect on systems performance, as well as receiver sensitivity.

In [Chapter 9](#), we lay out the fundamentals of lidar and address a variety of applications. Although each of these applications has a huge area of research behind it, there are nevertheless some common threads which we discuss in the context of our development. We also reference some relevant papers in these areas for those interested in further investigation.

In [Chapter 10](#), we discuss the performance of incoherent and coherent communications systems when operating in the optical turbulence channel. We will discuss a new statistical link budget approach for characterizing incoherent FSOC link performance, and compare experimental results with statistical predictions.

In [Chapter 11](#), we discuss a set of approaches for optical communications in diffusive scattering channel, augmenting the results described in [Chapter 9](#). Specifically, we highlight the accepted models for characterizing the statistical channel effects and techniques for providing communications in this channel.

As with many fields, we stand on the shoulders of great men and women whose contributions have shaped the field of engineering, especially optical system engineering. Of those researchers, the authors would like to acknowledge five pillars that have influenced the authors greatly through their teachings, personal interactions and insightful contributions – Professors Robert S. Kennedy, Irving S. Reed, Robert M. Gagliardi, Adolf Lohmann and Siebert Q. Duntley.

We also would like to thank our many colleagues who helped us over the years succeed in our optical systems projects to gain the insights we discuss in this book. They also have influenced us greatly. In writing this book, we want to acknowledge Drs. Larry Andrews, Juan Juarez, David Fried, Paul Kolodzy, Gary Lee, Ron Phillips, H. Alan Pike, David Young, and Mr. David Buck, Bob James, Todd Martin and Ned Plasson for their help and assistance.

In addition, we would like to acknowledge Dr. Anthony J. Tether, who over the years has encouraged and supported our various efforts in optical systems.

Finally, we recognize the biggest contributors to our book, our families, who have shown their patience during the preparation of this manuscript. Anyone who has imbedded themselves in an all-consuming goal knows the importance of the understanding and patience of loved ones.

Notation

$E[f(t)], \bar{f}$	expected value of $f(t)$
$m^2(t), \bar{m}^2$	modulation index; average value
$n(\vec{r}, t), \bar{n}$	Poisson rate parameter, average value
$2e^2 \bar{n} B, (2ei_{dc} B)$	referred to as shot noise
PWE	probability of symbol (word) error
(x, y)	coordinates of aperture plane
(x_0, y_0)	coordinates of receiver plane
$1/2B$	pulse width τ
a	absorption coefficient radius of the aperture, length of the aperture
A	area, optical amplitude
AO	adaptive optics
b	volume scattering coefficient, length of the aperture
B	bandwidth
$B(\mathbf{u})$	source irradiance function
BER	bit error rate
c, c_0	volume extinction coefficient; speed of light in free space
$C(\alpha), S(\alpha)$	Cornu spirals
C, C_0	information capacity (bits/s), constant, contrast; inherent contrast
C_{corr}	correlation length
C_n^2	refractive index structure parameter
C_w	curvature of the wave front
CNR	carrier-to-noise ratio
cx, cy	centroid in x- and y-direction
d	linear dimension of the photo-detector, aperture diameter
d_T, d_R	transmitter, receiver aperture diameter
d_S	beam diameter
d_d	photo-detector diameter
d_R	converging lense diameter
d_D	diffraction-limited focus diameter
$D; D_s$	mode number, dimension, aperture diameter; depth, Secchi depth
D_n	refractive index structure function
dB	decibel
dBmW	decibel milliwatts

e, q	electron charge
$E(z, r)$	spatial impulse response point spread function
E_d, E_u	downwelling, upwelling radiance at depth
E_b	energy per bit
erf	error function
erfc	complementary error function
F	F-number of lens
f	frequency (cycles/s)
$F(\mathbf{r}, t)$	solution to Fresnel integral (volts/m)
$\mathbf{f}, (f_x, f_y)$	spatial frequency, bold
f_c	focal length
f_g	Greenwood frequency
f_{sg}	normalized received power
F_w	off-axis irradiance correction
FEC	forward error correction
FOV	field of view
FSL	Fraunhofer spreading loss
G	gain
g	number of indistinguishable states in ground level of the atom, asymmetry factor
h	Planck's constant height
H	scale height
$h(\mathbf{r}, t)$	effect of single electron flow
$h(x, y; \omega)$	linear spatial effect at frequency ω
$H, E(I)$	irradiance (intensity) (watts/m ²)
HV	Hufnagel-Valley
HAP	Hufnagel-Andrews-Phillips
$I_e, I(\mathbf{R}); I(\mathbf{v})$	radiant intensity (watts/sr): image intensity 2D; image evolving in time
I_D	dark current
$I_0 (..)$	modified Bessel function of the first kind with order zero
J_0, J_1	zeroth and first-order Bessel functions
k	Boltzmann's constant (J/deg), electron count, diffuse attenuation coefficient
$\mathbf{k}, \kappa$	wavenumber vector
K	photo-electron counts
K	degrees Kelvin (degrees)
k_D, i_T	detection threshold
k_T	integer value of the detection threshold that minimizes the probability of error
ℓ_0	inner scale of turbulence
L_e	radiance (w/m ² sr), propagation path length
$L(z)$	normalized range equation (log of range corrected range equation $\times r^2$)
L_0	outer scale of turbulence
L_a, L_c	atmospheric, cloud transmittance
$L_{a/s}$	air/sea interface transmittance
m	mass of an atom

$M_e, M(\mathbf{r})$	radiant exitance (radiant emittance) (watts/m ²), point source coherence function, number of slots, number of symbols
M_{FM}	FM modulation index
M_L	number of local field modes
M_R	number of received modes
$m_v, m(t), m(\mathbf{r})$	Poisson intensity variable, number of electron counts; time modulation, spatial modulation
m_0	total noise counts
m_1	total signal plus noise counts
MTF	modulation transfer function
n	number of atoms; index of refraction of air
n_s	average number of photo-electrons per bit for a BER = 10 ⁻⁹
n_q	average number of photo-electrons per bit at the quantum limit
N_e	spectral radiance (w/m ² srλ); number density of air molecules, number of slots, number of Zernike modes
N_0	unilateral spectral noise density
NRZ	non-return-to-zero
OAGC	optical automatic gain control
OTF	optical transfer function
P	probability density
P_E, PE	probability of bit error
P_t	probability of t-errors
$P(x,y), P(z)$	pupil function; pressure
$p(\theta)$	scattering phase function
$P_e, P, P(z)$	radiant flux (radiant power) (watts); probability; pressure, power
PIB	power in the bucket
PIF	power in the fiber
POF	power out of the fiber
PRF	pulse repetition frequency
$Q(a,b)$	Marcum Q-function
Q, Q_e, Q_s	radiant energy (joules); extinction efficiency; scattering efficiency, compensation factor
R	distance, data rate, water reflectance
RZ	return to zero
$r, r(R_b)$	scalar range spectral efficiency, detector radius
$\vec{r}$	radius vector
r_c, r	radius, radius of the aperture
r_0	Fried parameter, radius of the aperture
$R(\rho)$	covariance of homogeneous process
$R(\tau)$	covariance of wide sense stationary process
R_n^m	radial polynomial
R_e	radius of the Earth
R_L	load resistor

RR	Rayleigh range
$\mathbf{r}, \mathbf{s}$,	vector distance (arrow, bold)
S	entropy (logarithm of degrees of uncertainty – information), spectral efficiency
$s, s(I, j)$	volume scattering coefficient, signal
$S(\omega)$	power spectral density
SNR, $[\frac{S}{N}]$	signal-to-noise ratio
SR	Strehl ratio
t, T, T_o	time, temperature
T_D	lase dead time
U	input potential to Fresnel integral (volts/m)
Var[]	variance
$V_{HV\ 5/7}$	HV 5/7 vertical profile for wind speed
VOA	variable optical attenuator
w	wind speed, error of focus
w_g	ground wind speed
w_s	beam slew rate
$W(x, y)$	aberration function
x	Mie parameter
$x(\mathbf{r}, t)$	shot noise process
z, R	range
Δ	maximum offset
ΔT	slotwidth
$\Delta\lambda$	spectral passband
$\Delta\nu$	Laser frequency linewidth
$\Phi_m(\mathbf{f}, t)$	spatial power spectral density of $m(\mathbf{r}, t)$
Φ_n	refractive index probability density function
$\Gamma(\mathbf{r}, \mathbf{r}'; t, t')$,	mutual coherence function
$M(\rho, z)$	
Λ	likelihood ratio
$\Theta()$	optical transfer function (OTF)
Ω	solid angle
α	detector sensitivity parameter
δ	Stephan-Boltzmann constant ($\text{J}\cdot\text{m}^2\ \text{deg}^{-4}\ \text{s}^{-1}$)
ε	emissivity
ε	focus error
$\varepsilon, \varepsilon_0$	permittivity, (free space)
φ	scalar potential, angular coordinate; beamwidth
$\varphi(\mathbf{p}), \varphi(\tau)$	eigenfunctions
Φ_c	laser signal phase angle
Φ_o	laser oscillator phase angle
φ	optical phase
η	quantum efficiency
γ_{Tx}, γ_{Rx}	transmitter, receiver transmittance

γ_{fiber}	fiber transmittance
γ_0	signal-to-noise ratio per symbol per unit bandwidth
$I(n, x)$	incomplete gamma function
λ	wavelength
$\lambda_{\tau, l}, \lambda_{s, j}, \lambda(q, l)$	eigenvalues
μ, μ_0	permeability, (free space)
$\mu_{S, B}$	average number of photo-electrons per bit from laser signal
$\mu_{N, B}$	average number of photo- electrons per bit from background radiation and dark current
$\pi_k(\theta), \xi_k(\theta)$	spherical harmonics (can be written as Legendre polynomials)
$\theta, \alpha, \varphi_s$	angular coordinate
θ_0	isoplanatic angle
ρ	reflection coefficient; bits of resolution
ρ_0	plane wave lateral coherence length
σ	cross section, standard deviation
σ_{fit}^2	residual fitting error
σ_χ^2	Rytov number
σ_T^2	angle of arrival tilt variance (jitter)
σ_θ^2	tilt variance
σ_{iTH}^2	thermal noise variance
τ	optical thickness, time between symbol errors
τ_d	diffusion thickness
τ_0	Greenwood time period
τ_B	bit period
ω	frequency (rad/s); single-scatter albedo, radian frequency
ψ, ψ_{BB}	Wien's law, black-body equation

1 Genesis of electro-optic systems

When we decided to write this book about the design of electro-optic systems, we agreed to make it as fundamental as possible. To do this in detail would most probably make the book unwieldy. Rather, we will try to motivate all aspects of the design from fundamental principles, stating the important results, and leaving the derivations to references. We will take as our starting point the first two Laws of Thermodynamics [1]. The Three Laws of Thermodynamics are the basic foundation of our understanding of how the Universe works. Everything, no matter how large or small, is subject to the Three Laws. The Laws of Thermodynamics dictate the specifics for the movement of heat and work, both natural and man-made. The First Law of Thermodynamics is a statement of the conservation of energy – the Second Law is a statement about the nature of that conservation – and the Third Law is a statement about reaching Absolute Zero (0° K). These laws and Maxwell's equations were developed in the nineteenth century, and are the foundation upon which twentieth-century physics was founded.

Since the Laws are so obviously important, what are they?

First Law: energy can neither be created nor destroyed. It can only change forms. In any process, the total energy of the universe remains the same.

The First Law states that energy cannot be created or destroyed; rather, the amount of energy lost in a steady-state process cannot be greater than the amount of energy gained. This was a rebuff to early attempts at perpetual motion machines. This is the statement of conservation of energy for a thermodynamic system. It refers to the two ways that a closed system transfers energy to and from its surroundings – by the process of heating (or cooling) and the process of mechanical work. Whether the system is open or closed, all energy transfers must be in balance.

Second Law: the entropy of an isolated system not in equilibrium will tend to increase over time, approaching a maximum value at equilibrium.

The second law states that energy systems will always increase their entropy rather than decrease it. In other words heat can spontaneously flow from a higher-temperature region to a lower-temperature region, but not the reverse. (Heat can be made to flow from cold to hot, as in a refrigerator, but requires an external source of power, electricity.)

A way of looking at the second law for non-scientists is to look at entropy as a measure of disorder. A broken cup has less order than before being broken. Similarly solid crystals have very low entropy values, while gases which are highly disorganized have high entropy values.

1.1 Energy

What does this have to do with optics? Well, about the same time that the thermodynamic laws were being developed, the laws of electricity were also being developed. James Clerk Maxwell unified the Laws of Gauss, Ampere and Faraday into the set of laws now referred to as Maxwell's equations [2]. These equations show that electric and magnetic fields are coupled and satisfy a wave equation,

$$\nabla^2 \phi = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} \quad (1.1)$$

where ϕ can represent the scalar components of the electric and magnetic fields. In a vacuum the speed of light $c = 1/\sqrt{\epsilon_0 \mu_0}$ where ϵ_0, μ_0 are the permittivity and permeability of free space, as derived in Maxwell's equations. Since electromagnetic fields can propagate in a vacuum, no loss occurs (unless obstacles are encountered) and the fields can propagate indefinitely since energy can be neither created nor destroyed. Thus the energy from the Sun can bring life to the planet Earth, and we can see the vast vista of stars at night, even though they are light years away. A general solution to the wave equation (in spherical coordinates) for the fields takes the form

$$\phi = \text{Constant} \times \frac{f\left(t - \frac{r}{c}\right)}{r}. \quad (1.2)$$

$t - \frac{r}{c}$ is referred to as the retarded time. That is, the waveform of the field propagates undistorted, but attenuated (spread out in space) by the factor r , when traveling the distance r , in a time r/c . Since the energy in the fields is proportional to $|\phi|^2$ [2], we see that the energy in a propagating electromagnetic field falls off as $1/|r|^2$. From this we can deduce several properties regarding the transfer of energy by electromagnetic means. For the purpose of this book we will use the following standard notation for Radiant quantities:

Radiant energy is energy carried from any electromagnetic field. It is denoted by Q_e .

Its **SI** unit is the **joule** (J).

Radiant flux is radiant energy per unit time (also called **radiant power**); it is considered the fundamental radiometric unit. It is denoted by P_e . Its **SI** unit is the **watt** (W).

Radiant exitance, or **radiant emittance**, is radiant flux emitted from an extended source per unit **source area**. It is denoted by M_e . Its **SI** unit is: watt per square meter (W/m^2).

Irradiance is radiant flux incident on a surface unit area. It is denoted by E_e . Its **SI** unit is: **watt per unit area** (W/m^2).

Radiant intensity is radiant flux emitted from a point source per unit **solid angle**. It is denoted by I_e . Its **SI** unit is: watt per **steradian** (W/sr).

Radiance is radiant flux emitted from an extended source per unit **solid angle** and per unit **projected source area**. It is denoted by L_e . Its **SI** unit is: watt per steradian and square meter ($\text{W}/(\text{sr m}^2)$).

Spectral in front of any of these quantities implies the same **SI** unit per unit **wavelength**.

Thus spectral radiant energy is the energy radiated per unit wavelength. These noise sources generally are not described analytically, and depending upon the instrument used fall into one of the categories described above.

1.2 The range equation

We will try to show how these terms are used, with a few examples.

Example 1.1 Suppose a point source radiated (transmitted) an amount of radiant power, $P_t = Q_e$ joules/second, equally into all directions (4π steradians or omnidirectional). Then the radiant intensity would be equal to

$$I_t = \frac{P_t}{4\pi} \quad (1.3)$$

A receiver with an area A_r located a distance r from the point source would subtend the solid angle $A_r/|r|^2$ to the source. Therefore such a receiver could capture an amount of radiant power (power) equal to Figure 1.1,

$$P_r = \frac{P_t A_r}{4\pi |r|^2} \quad (1.4)$$

which conforms to the $1/|r|^2$ requirement of a propagating field. Suppose that a second identical source could direct all its energy toward a receiver, uniformly into a solid angle $\Omega < 4\pi$. Such a source would then transmit $4\pi/\Omega$ times more power to the receiver, without the receiver knowing that it wasn't a source with $(4\pi/\Omega)P_t$ watts. For this reason $4\pi/\Omega$ is called the gain (G_t —over isotropic) of the second system, and we write the received power as

$$P_r = \frac{P_t A_r G_t}{4\pi |r|^2} \quad (1.5)$$

This is referred to as the frequency-independent form of the range equation, and defines the transfer of power in a radiating system.

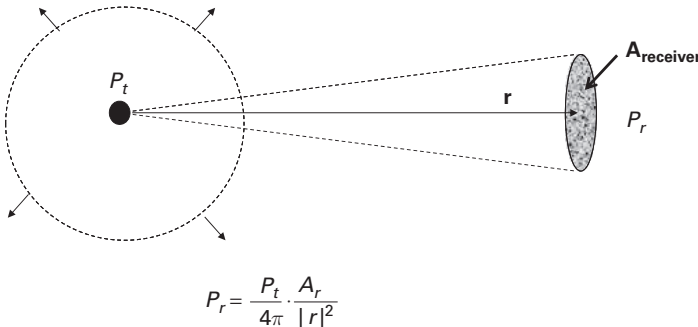


Figure 1.1 Range equation in geometric form.

Example 1.2 How can we calculate the total power emitted by the Sun? The Sun is located a distance $R_{sun} = 92 \times 10^6$ miles from the Earth. The irradiance of the Sun at the top of the Earth's atmosphere (solar constant), I_{sun} , is approximately 1500 w/m^2 . The Earth has a cross section, A_{earth} , of $\pi(4000)^2 = 50.3 \times 10^6$ square miles, or a cross section of $1.3 \times 10^{14} \text{ m}^2$. Thus the Earth intercepts $I_{sun}A_{earth}$ watts from the Sun, which is approximately 1.96×10^{17} watts. The solid angle subtended from the Earth to the Sun is $\Omega_{sun-earth} = A_{earth}/R_{sun}^2$ steradians, so the Earth intercepts the percentage $\Omega_{sun-earth}/4\pi$ of the Sun's emission, or only 5.81×10^{-7} per cent of the energy radiated by the sun. This means that the sun emits a total of $(4\pi/\Omega_{sun-earth})I_{sun}A_{earth} = 3.37 \times 10^{25}$ watts.

1.3 Characterization of noise

First we will relate all the radiant quantities described earlier. We start first with the most elemental of the quantities, the spectral radiance, $N_e(r, \theta, \lambda)$. To obtain the radiance, we would perform the integration, which for narrowband filters can be approximated, as follows:

$$L_e(r, \Omega) = \int_{\lambda_1}^{\lambda_2} N_e(r, \Omega, \lambda) d\lambda \approx N_e(r, \Omega, \lambda) \Delta\lambda \quad (1.6)$$

From $L_e(r, \Omega)$ we can go in one of two directions. To obtain the radiant emittance we perform the integral

$$M_e(r) = \int_0^{\Omega_0} L_e(r, \Omega) d\Omega \approx L_e(r, \Omega) \Omega_0 = N_e(r, \Omega, \lambda) \Omega_0 \Delta\lambda \quad (1.7)$$

while to get the radiant intensity, we perform the integral

$$I_e(\Omega) = \int_A L_e(r, \Omega) dS \approx L_e(r, \Omega) A = N_e(r, \Omega, \lambda) A \Delta\lambda \quad (1.8)$$

Finally, to obtain the power, we perform either one of two integrals:

$$P_e = \int_A M_e(r) dS \approx M_e(r) A = L_e(r, \Omega) A \Omega_0 = N_e(r, \Omega, \lambda) A \Omega_0 \Delta\lambda \quad (1.9)$$

or

$$P_e = \int_0^{\Omega_0} I_e(\Omega) d\Omega \approx I_e(\Omega) \Omega_0 = L_e(r, \Omega) A \Omega_0 = N_e(r, \Omega, \lambda) A \Omega_0 \Delta\lambda \quad (1.10)$$

We have defined I_e to be from a point source, but by allowing a variation with angle it becomes more general. A point source would have no variation with angle. We have also used Ω to denote (θ, ϕ) and, $d\Omega$ the Jacobian. The Sun, which has a diameter of 863 000 miles, only subtends 0.5 degrees to Earth, and for many cases appears like a point source, but can project different values in different directions. We have used the description of frequency and bandwidth in customary terms of wavelength and differential wavelength, because these sources were first studied by astronomers, and that is

how they were described. We also have the relationship between frequency and wavelength of $f = c/\lambda$ with $df = |(c/\lambda^2)|d\lambda$.

Example 1.3 Let us assume that a source radiates a spectral radiance $N_\lambda(w/m^2 - sr - BW)$. Suppose that the receiver has a field of view (FOV – the solid angle that it views) equal to Ω_{rec} , a collection area A_{rec} , and a filter bandwidth $\Delta\lambda$. If R is the range from the source to the receiver, then integrating over the source area yields (1) $\Omega_{rec}R^2$ if the source fills the FOV, or (2) A_{source} if it doesn't. The receiver subtends the solid angle A_{rec}/R^2 to the source. Therefore the total power collected by the receiver becomes

$$N_\lambda \Omega_{rec} R^2 \frac{A_{rec}}{R^2} \Delta\lambda = N_\lambda \Omega_{rec} A_{rec} \Delta\lambda \quad (1.11)$$

if the source fills the receiver FOV. This would be in cases like imaging systems, where the FOV of each detector pixel fills the cell in the image that is being viewed. On the other hand in a detection system, such as a system searching for a hot target, the FOV might be larger than the target and

$$N_\lambda \frac{A_{source}}{R^2} A_{rec} \Delta\lambda \quad (1.12)$$

would most likely apply.

1.4 Black-body radiation

Black-body or “pure-temperature” radiation was the name given to radiation emanating from a system in thermal equilibrium. This is the fundamental source of primary (suns, stars, plasmas, etc.) and secondary (moons, planets, atmosphere, etc.) electromagnetic noise, and was the exciting area of research in thermodynamics. In addition, the outcome proved fundamental to the development of the quantum theory.

It was known that in a sealed chamber at a given temperature, the spectral distribution of radiation is independent of the material out of which the chamber is made. In 1884, Boltzmann [3] showed that the total irradiance energy in this distribution varied as the fourth power of the temperature. The constant of proportionality, $\delta = 0.567 \times 10^{-8} \text{ J-m}^{-2} \text{ deg}^{-4} \text{ s}^{-1}$, is known as the Stefan-Boltzmann constant,

$$\text{Irradiance} \sim \delta T_0^4 \quad (1.13)$$

and the law as the Stefan-Boltzmann law. In 1893 Wien [4] showed that this functional dependence must be of the form

$$\psi_\lambda \sim T_0^5 f(\lambda T_0) = \frac{1}{\lambda^5} F(\lambda T_0) \quad (1.14)$$

where T_0 is the temperature in degrees Kelvin. He assumed a form

$$\psi_\lambda \sim \lambda^{-5} e^{k/\lambda T_0} \quad (1.15)$$

called Wien's law. He also developed Wien's displacement law, which states that if the product λT_0 is held constant, ψ_λ / T_0^5 is the same at all temperatures.

In 1900, Rayleigh [5] made a suggestion which was based on a technique used successfully in statistical mechanics. This was followed up with a calculation made by Jeans, resulting in the Rayleigh-Jeans formula. The idea relied on the equipartition of energy, and went as follows: one can expand the electromagnetic field contained in an enclosure into orthogonal and independent modes (degrees of freedom). By then associating the total energy kT_0 with each degree of freedom (used in the ideal gas law) an expression for the radiation can be obtained. Jeans calculated the number of modes per cycle to be D_{BB} , where

$$D_{BB} = \frac{8\pi f^2}{c^3} \quad (1.16)$$

yielding the expression for the energy to be

$$E = kT_0 D_{BB} = \frac{8\pi f^2 kT_0}{c^3}. \quad (1.17)$$

Unfortunately, this predicted an infinite energy instead of the fourth-power dependence on T_0 . The two expressions E and ψ_λ , are plotted in Figure 1.2, along with precise

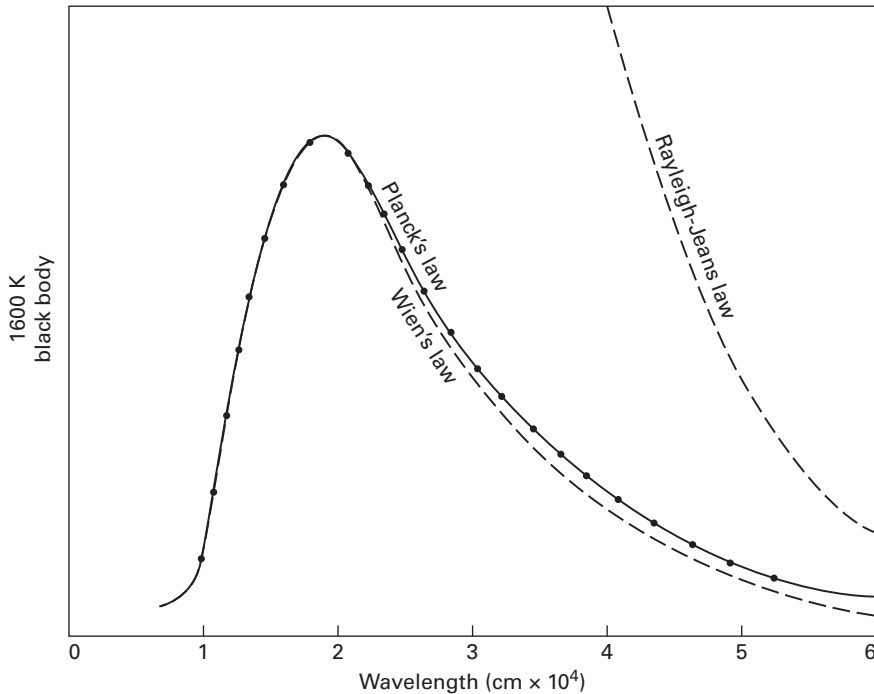


Figure 1.2 Black-body radiation, Wien's law, and the Rayleigh-Jeans approximation. The experimental curve is fit exactly by Planck's law.

measurements taken by Lummer and Pringsheim [6]. Neither was correct, but Wien's law was asymptotically correct at high frequencies, and the Rayleigh-Jeans law was asymptotically correct at low frequencies.

This was the picture as seen by Planck around 1900. Planck's derivation of the correct equation for the black-body law came through his understanding of thermodynamics. He curve-fitted the second derivative of entropy with respect to energy to yield the correct two asymptotic forms, and worked them back to obtain the energy equation [7]. The most commonly used derivation (suggested by Einstein) multiplies the mode density D_{BB} by the quantum mechanical equivalent of kT_0 , which is

$$\frac{hf}{e^{hf/kT_0} - 1} \quad (1.18)$$

Thus we see that the equation for black-body radiation can be derived to be

$$\begin{aligned} \psi_\lambda &= \frac{8\pi f^2}{c^3} \frac{hf}{e^{hf/kT_0} - 1} \\ &= \frac{8\pi hc}{(\lambda)^5} \frac{1}{e^{hc/k\lambda T_0} - 1} \end{aligned} \quad (1.19)$$

which fits the energy density curves exactly.

This is the energy per unit of wavelength (bandwidth) contained in a sealed container. When we consider the radiation from an isotropic black body, we must multiply the energy by the speed of light, c , divided by 4π , for isotropy, and divide by two for each polarization. This yields

$$\psi_\lambda \frac{c}{4\pi} \frac{1}{2} = \frac{8\pi f^2}{c^3} \frac{hf}{e^{hf/kT_0} - 1} \frac{c}{4\pi} \frac{1}{2} = \frac{f^2}{c^2} \frac{hf}{e^{hf/kT_0} - 1} \quad (1.20)$$

which has the dimensions of energy per unit bandwidth per unit solid angle per unit area. As such it is a spectral radiance, N_{BB} . Reflecting back on example 1.3, the Sun appears as a 5778° black body. If the receiver field of view is greater than 0.5 degrees we can treat example 1.2 as before (example 1.3b). If the field of view is less than 0.5 degrees we are only looking at a small cell in the Sun, which in turn fills the field of view of the pixel viewing it (example 1.3a). When an instrument measures the solar irradiance, E_{sun} , its field of view is larger than that subtended by the Sun and hence measures

$$E_{sun} = N_\lambda \frac{A_{sun}}{R^2}, \quad (1.21)$$

which has dimensions of joules/ μm^2 as given in example 1.2.

The quantum mechanical equipartition energy has the two asymptotic forms. At the low energy (low frequency), $hf < kT_0$, we have

$$\frac{hf}{e^{hf/kT_0} - 1} \approx \frac{hf}{1 + \frac{hf}{kT_0} - 1} = kT_0 \quad (1.22)$$

the classic value. At the high end we have $hf \exp(-hf/kT_0)$, which approaches Wien's law.

What Planck also observed about black-body radiation was that this described a sum of harmonic oscillators,

$$\frac{8\pi f^2}{c^3} \frac{hf}{e^{hf/kT_0} - 1} = \frac{8\pi f^2}{c^3} \frac{hf}{e^{hf/kT_0} \left(1 - e^{-hf/kT_0}\right)} = \frac{8\pi hf^3}{c^3 e^{hf/kT_0}} \sum_{n=0}^{\infty} e^{-nhf/kT_0} \quad (1.23)$$

which was the first quantum hypothesis.

We have expanded on the derivation of black-body radiation because it is so important in understanding the composition of background noise. This will become clearer as we see the relevance of modes, mode number and energy per mode.

We show several representative curves of astronomical radiation in Figures 1.3–1.10. In Figure 1.3, we show the approximate black-body radiation of the Sun outside the atmosphere, and also modified by the absorption bands of the atmosphere [8]. In Figure 1.4, we show this in logarithmic form as a function of the zenith angle. Notice that the units are different, but the results are the same [8]. In Figure 1.5, we show the sky radiance during the day, which

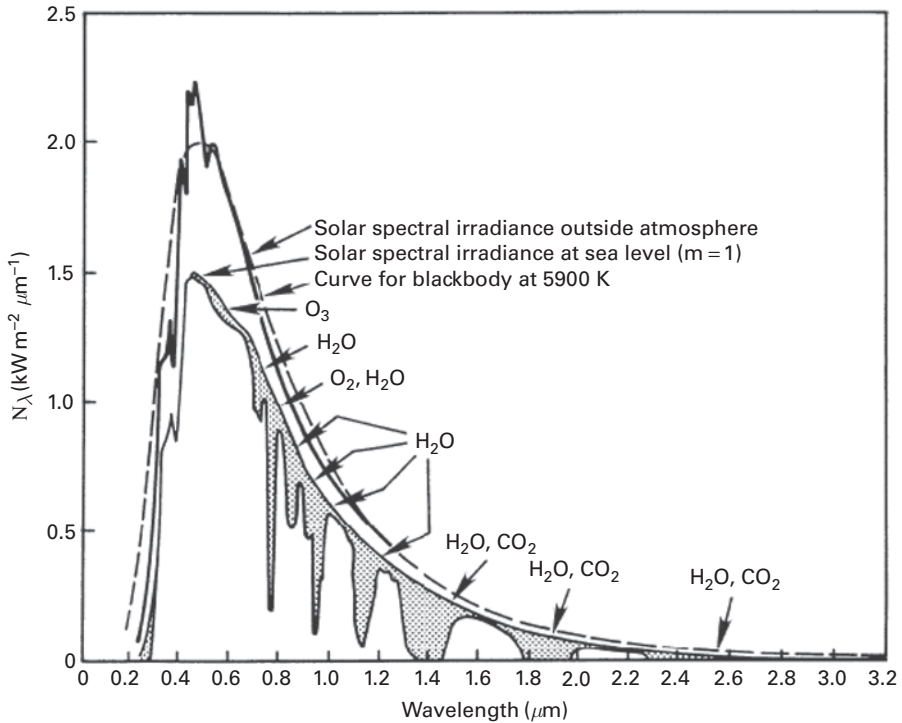


Figure 1.3 Spectral distribution curves related to the Sun. The shaded areas indicated absorption at sea level due to the atmospheric constituents shown.

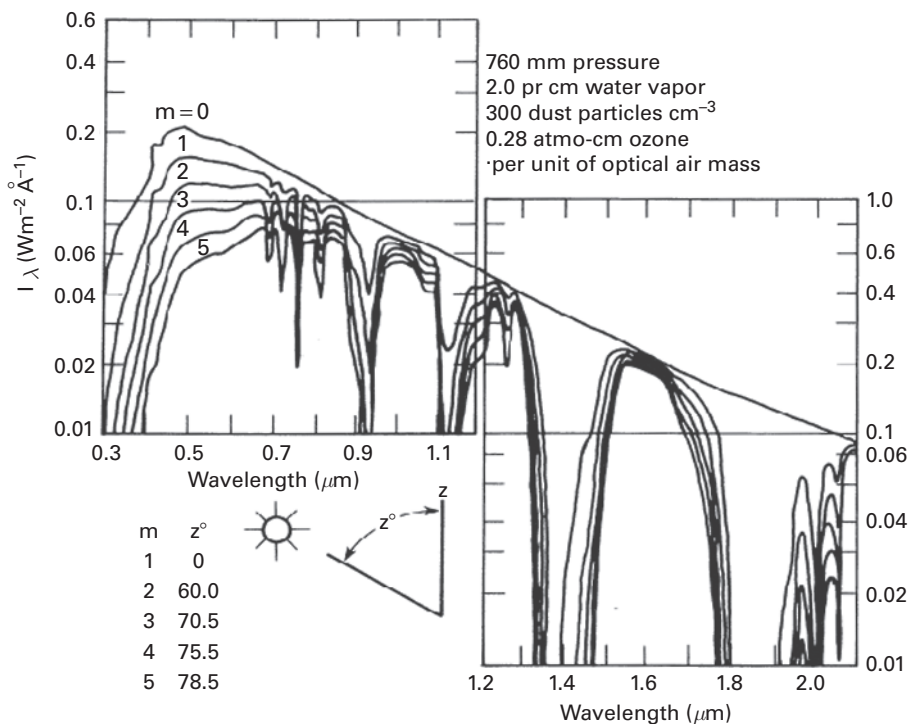


Figure 1.4 Solar spectral irradiance curves at sea level for various optical air masses. The value of the solar constant used in this calculation was 1322 W/m^2 [23].

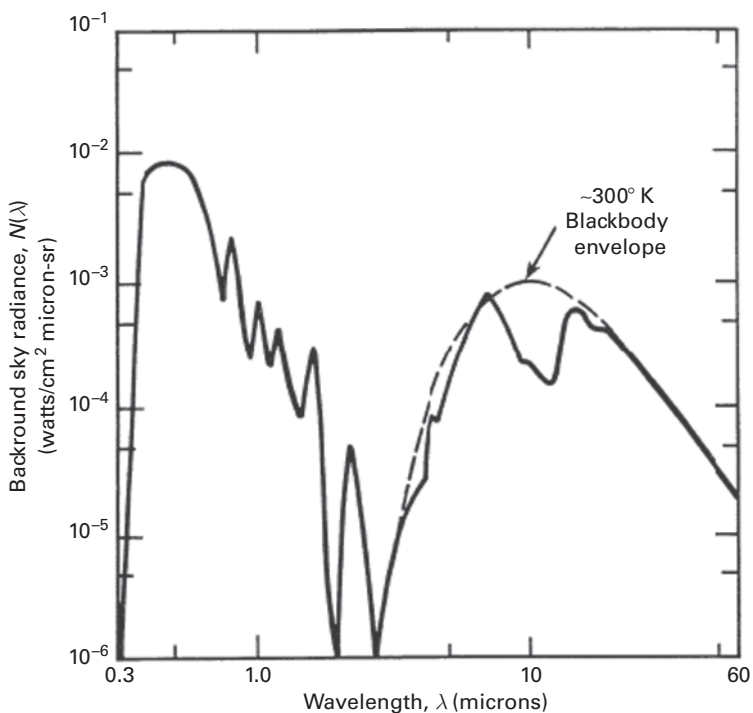


Figure 1.5 Diffuse component of typical background radiance from sea level; zenith angle, 45° ; excellent visibility [23].

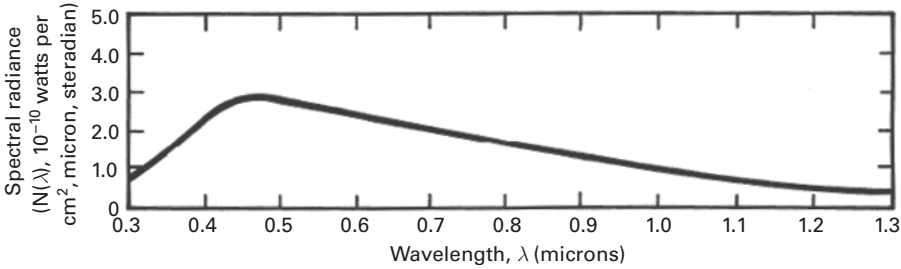


Figure 1.6 Nighttime sky radiance from zenith due to zodiacal light, galactic light, and scattered starlight [23]. (Courtesy of National Bureau of Standards.)

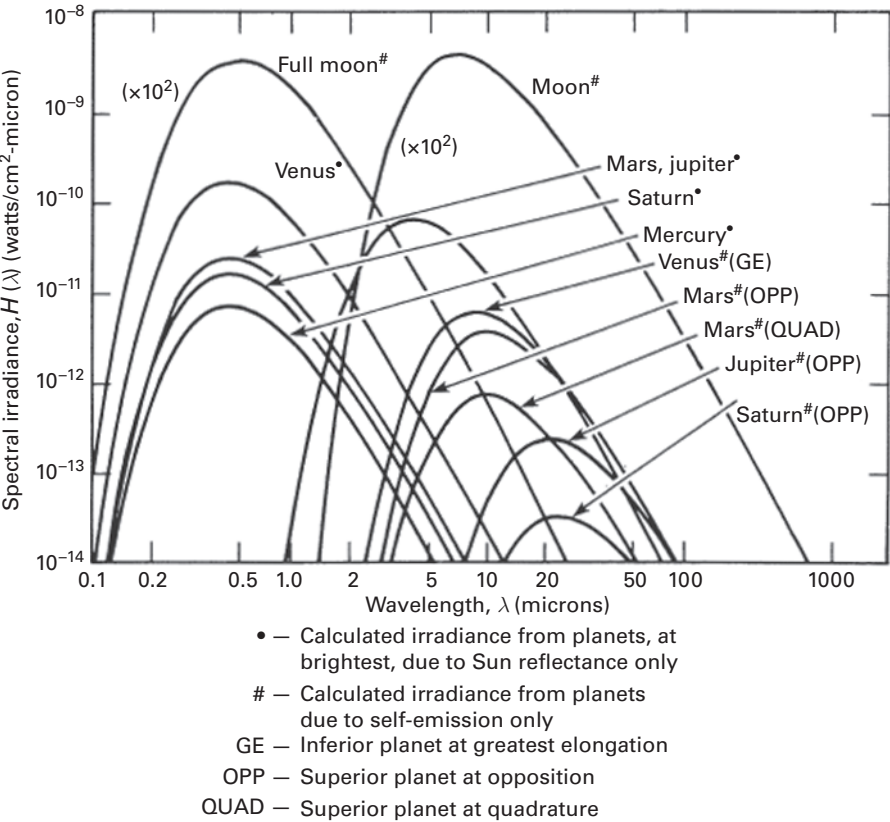


Figure 1.7 Calculated planetary and lunar spectral irradiance outside the terrestrial atmosphere [23].

consists of the re-radiation of the solar energy plus the re-radiation of the Earth's emission at approximately 300° Kelvin [9]. In Figure 1.6, we see the sky radiance at night [10]. In Figure 1.7, we show the irradiance from the planets, Moon and Venus [11]. In Figure 1.8, we see their spectral albedo (spectral albedo is defined as the ratio of the total reflected energy

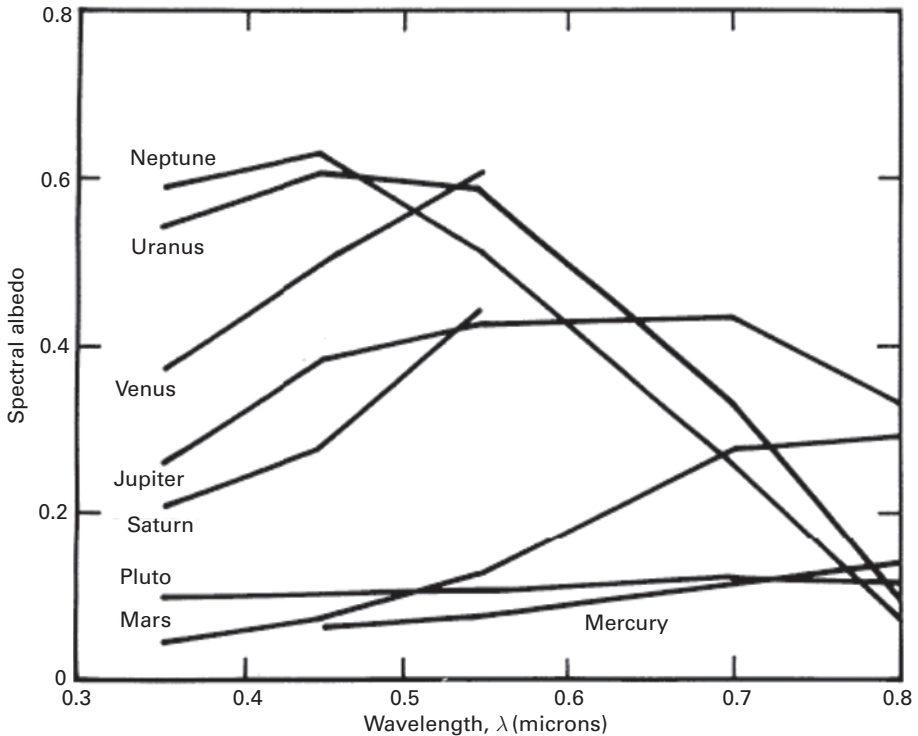


Figure 1.8 Spectral albedo as a function of wavelength [23].

from a surface to the total incident energy on said surface) [12]. In [Figure 1.9](#), we see the spectral radiant emittance of the Earth as viewed from space [13]. Finally in [Figure 1.10](#) we see the spectral irradiance of the brightest stars. A further discussion of these curves can be found in [14]. For an intensive description of the development of the black-body theory and its link to the development of quantum mechanics see Whittaker [15].

1.5 Entropy

As the Second Law shows, systems tend towards a balance of disorder, with entropy always increasing. Thus if all air conditioning and heating were turned off in a house, the temperature would eventually equal that of its surroundings. While it is possible to build a machine that can raise (or lower) the house temperature, these mechanical systems can at best make a perfect exchange of heat. In fact, all mechanical systems have some inefficiency; hence some waste energy is generated. (And in fact, the universe is constantly losing usable energy.)

Entropy is defined as the measure of uncertainty in a system. For the case of an ideal monatomic gas in a container of volume V that is thermally insulated from its surroundings, the entropy can be expressed using the Sackur-Tetrode equation [16] by

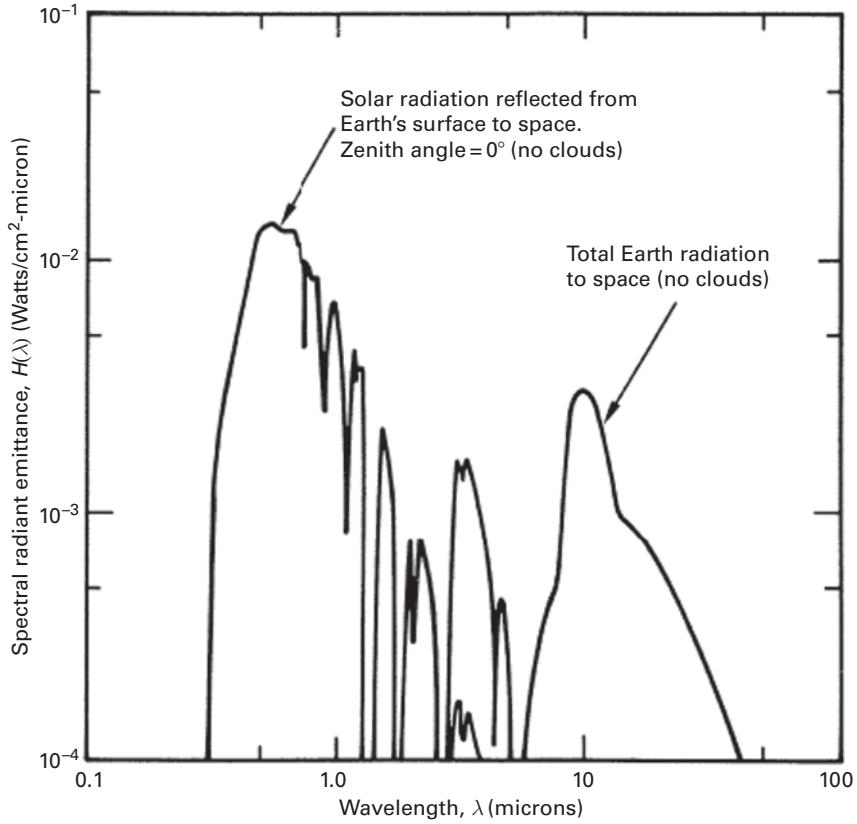


Figure 1.9 Spectral radiant emittance of the earth [23].

$$S = kn \left[\frac{5}{2} + \ln \left(g \frac{V}{n} \left\{ \frac{4\pi m E}{3h^2 n} \right\}^{3/2} \right) \right] \quad (1.24)$$

where

n = number of atoms

m = mass of an atom

k = Boltzmann's constant

h = Planck's constant

E = total energy

g = number of indistinguishable states in the ground level of the atom.

If the ground state is not degenerate, $g = 1$. This is the case of an atom with no momentum in the ground state ($j = 0$). If there is a moment of momentum j (spin plus orbit), there are $g_j = 2j + 1$ states, which are supposed to be equally probable. If we have spin $j = 1/2$ (two orientations), then $g = 2$. For $j = 1$, $g = 3$ and three orientations ($-1, 0, +1$). If we compute S_1 for $g = 1$, then $S_2 = S_1 + kn(\ln 2)$ and $S_3 = S_1 + kn(\ln 3)$. The importance of this example is that we see the entropy is proportional to the logarithm of the possible orientations of the atom.

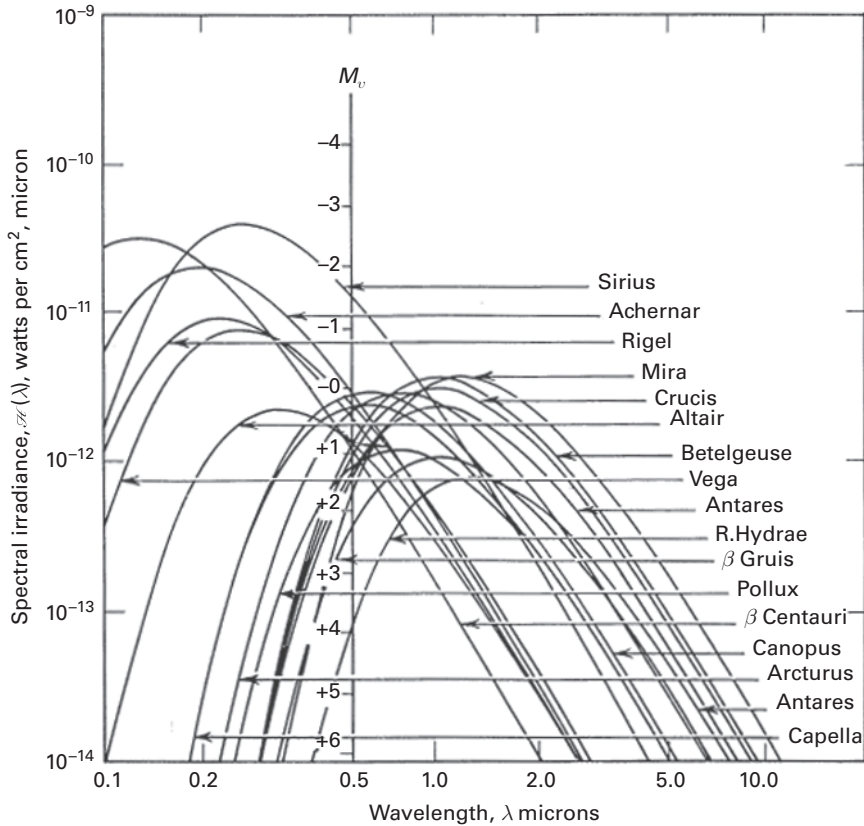


Figure 1.10 Spectral irradiance of brightest stars outside the terrestrial atmosphere [24].

Information too can be written as the logarithm of the total amount of uncertainty. This idea had been used by Nyquist [17] and Hartley [18].

Example 1.4 Suppose we had to guess a number that existed between zero and 31 ($32 = 2^5$ possibilities), and suppose we could ask n yes/no questions. How many questions would it take to find the number? The first question should be “is it between zero and 15”? If the answer is yes, we ask whether the number is between zero and seven. If the answer is no, we ask whether the number is between 8 and 11. And so forth. If, for example, the number was 27 we would get a no first, a no second, a yes third and two more no’s. If we assign zero to a yes and a one to a no, we would get 11011, or the binary word for 27. Thus we see that information is the negative of entropy [19]. If we assigned one bit of information to each binary digit we would have five bits of information. We see that the total amount of information would be $\log_2(32) = 5$ bits; Figure 1.11. If we were to pick two numbers independently between zero and 31, we would now have $32 \times 32 = 1024$ possibilities. Since $\log_2(1024) = 10 = 5 + 5$, we see that information is additive as it should be.